Kursöversikt för Computer Engineering MA, Signal and Image Processing DT081A D3055 HT2026

Kursplan (DT081A) (https://www.miun.se/utbildning/kursplaner-och-utbildningsplaner/DT081A/)

Välkomstbrev (D3055) (https://www.miun.se/student/antagen/valkomstbrev/show?vbKod=HT2026_DT081A_D3055)

Hoppa fram till i dag

▶ Course Responsible Teacher

▶ Teaching Assistant

▶ Faculty Administrator

Welcome to this course! Here is an overview:

Aim and learning objectives

Course Overview:

In this 6 ECTS credit course, you will explore the theoretical foundations and practical applications of signal and image processing techniques. The course covers a wide range of topics, from basic signal representations to advanced AI-based processing methods. You will learn to analyse, design, and implement various processing algorithms, preparing you for cutting-edge work in fields such as computer vision, image analysis, and machine learning for visual media.

Key Topics:

- Signal and system representations
- Time-domain and frequency-domain analysis
- Digital filter design
- Spatial-domain image processing
- Applications of signal and image processing in visual AI

Course objectives

Upon completing the course, the student will be able to:

- describe and explain the theoretical fundamental principles within signal and image processing
- apply the fundamental principles of signal and image processing in both traditional methods and modern AI-based systems.
- select and apply appropriate signal and image analysis techniques on specific problems with visual media.
- describe and explain algorithms within signal and image processing based on mathematical foundations.

- design and implement filters for signals and images in both time, space, and frequency domains.
- evaluate and critically analyze the performances of different signal and image processing algorithms, using tools and programming languages relevant to the area.

Entry requirements

- Computer Engineering, 60 credits including
  - Object oriented programming, 15 credits
- Mathematical subjects, 30 credits including
  - Probability theory and statistics
  - Linear algebra

Teaching and learning methods

Course Structure:

- 12 Lectures
- 6 Exercise sessions
- 3 Laboratory exercises
- Self-study materials and assignments

Lectures

There are 12 lectures of 2 x 45 min. The lectures are a resource for you and are a complement to the textbooks. During the lectures, I present the main topics of the course and the important aspects, but you must also read the referenced literature to cover the entire material. Before attending the lecture, you should review the corresponding theory sections in the book. After listening to the lecture, you should read the book in more detail.

Exercises

There are 6 exercises of 2 x 45 min. There are a number of practical tasks to complete, which give you additional opportunities to absorb the course content. They are also necessary to pass the exam. Review the tasks before the teacher-led exercise session.

Laboratory sessions

There are 3 laboratory sessions of 4 hours. The labs are designed to give you a practical application of the knowledge you have learnt in the lectures. The labs are carried out in a room where computers are available to solve the problems. You may also use your own computer. Prepare the labs well in order to complete them on time by reading the corresponding theory and the lab instructions. A supervisor will be available in the lab room.

Working in group

Another important part of the learning process is working in a group. By discussing, asking and explaining to your fellow students, you get alternative perspectives and also good practice in expressing the course content in your own words. You will also remember the problem and its solution better. It is therefore important that you are part of a study group where you, for example, solve the exercises together and discuss lab assignments.

Examinations

Read more on the page ***Course examinations*** **(https://canvas.miun.se/courses/12673/pages/about-the-course-examinations)**

Course literature

Read more on the page ***Course literature*** **(https://canvas.miun.se/courses/12673/pages/about-the-course-literature)**

Instructions for generative AI in the course

In connection with studies at university, there is often talk of cheating with the help of AI, but AI can also be a great benefit in your studies. Instructions for generative AI in the course is described on the page **Instructions for generative AI in the course (https://canvas.miun.se/courses/12673/pages/instructions-for-generative-ai-in-the-course)**

Kurssammanfattning:

Datum	Information	Sista inlämningsdatum
	Lab 1 - Introduction (https://canvas.miun.se/courses/12673/assignments/46039)	
	Lab 2 - Time and Frequency Domain (https://canvas.miun.se/courses/12673/assignments/46040)	
	Lab 3 - Image Processing (https://canvas.miun.se/courses/12673/assignments/46041)	